you can make necessary adjustments. For example, "Test Problem: What is the area if r = 5?"

5 *Iterate and refine.* Let's say that the preceding steps revealed that the results were inconsistent. In that case, you may need to refine the prompt to improve clarity, address any issues identified, and further encourage the desired reasoning process. For example, "Refined Prompt: Calculate the area of a circle with r = 5." Here we used explicit wording to eliminate any confusion about what shape we are referring to and remove the need for the model to guess the type of the shape from the question.

To recap, chain-of-thought prompting enhances the reasoning capabilities of LLMs by guiding them to present each individual step involved in solving a specific problem. When implemented effectively, this technique can significantly boost the model's performance with less effort.

Now we will explore the concept of embeddings and how they can be effectively utilized within the Vercel AI SDK. Embeddings serve as a powerful tool for representing words, phrases, or images as vectors in a high-dimensional space, enabling applications such as semantic search and retrieval-augmented generation, which I expand on in chapter 7.

## 5.4 Embeddings: Giving AI a sense of meaning

When a user inputs a prompt, the AI model converts the words into embeddings. Each word or token in the prompt is transformed into a vector that reflects its meaning and context within the sentence. The primary purpose of embeddings is to encapsulate individual measurable properties of the data, allowing AI models to use these features for tasks such as recommendations, classifications, and similarity searches.

Embeddings are thus numerical representations of data that effectively capture relationships and patterns within complex datasets. Processing various types of input, such as plain text, images, or audio, AI models require this data to be transformed into low-dimensional vectors (tuples of floating-point numbers). This transformation simplifies the data while retaining its essential features, resulting in embeddings.

### 5.4.1 The restaurant menu analogy: A taste of embeddings

Imagine you run a restaurant and a customer asks, "Can you recommend a dish similar to spaghetti carbonara?" The human staff knows that means they like creamy pasta dishes. The staff doesn't have to scan every ingredient of every dish; they can intuitively search for similar concepts.

This is what embeddings do for AI. The "customer order" is the user's query, which is converted into a vector. The "menu items" are also precomputed vectors. The system then performs a similarity search to find the closest menu items in the embedding space. The "attributes" (like "pasta" or "creamy") are the dimensions that define the vectors, capturing the essential qualities of each dish. This process allows for fast, relevant recommendations without the need for complex keyword matching. Figure 5.8 shows the main components of embeddings using a restaurant menu analogy.